

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 5 Solutions

1. (a) Let y_1 be the amount (in gram) of chlorine in tank 1 at time t . So $y_1(0) = 150$ g.
Let y_2 be the amount (in gram) of chlorine in tank 2 at time t . So $y_2(0) = 50$ g.
Then

$$\begin{aligned} y_1' &= -\frac{y_1}{20} \text{ g/l} \times 6 \text{ l/min} + \frac{y_2}{10} \text{ g/l} \times 3 \text{ l/min} + 0 \text{ g/l} \times 3 \text{ l/min} \\ y_2' &= \frac{y_1}{20} \text{ g/l} \times 4 \text{ l/min} - \frac{y_2}{10} \text{ g/l} \times 4 \text{ l/min} \end{aligned}$$

So the underlying matrix of this SDE is $\begin{pmatrix} -0.3 & 0.3 \\ 0.2 & -0.4 \end{pmatrix}$.

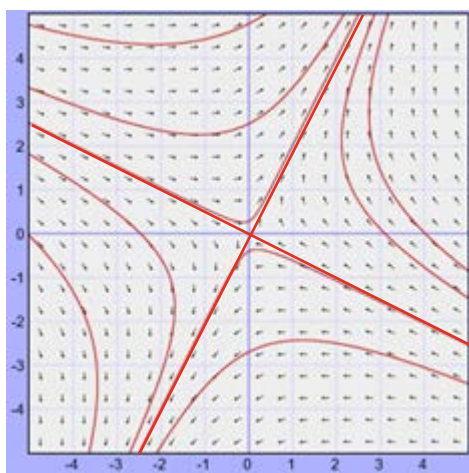
- (b) The eigenvalues can be calculated as -0.1 and -0.6 with corresponding eigenvectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

So the general solution of the system is $c_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{-t/10} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t/5}$.

With the initial condition, we have $3c_1 + c_2 = 150$ and $2c_1 - c_2 = 50$. This gives $c_1 = 40$ and $c_2 = 30$.

So the amount of chlorine in tank 1 is $y_1 = 120e^{-t/10} + 30e^{-3t/5}$ and in tank 2 is $y_2 = 80e^{-t/10} - 30e^{-3t/5}$.

- (c) Since both eigenvalues are negative, the two exponential functions are decreasing and become 0 in the long run (when $t \rightarrow \infty$).
2. (a) The eigenvalues of $\mathbf{A}$ are 2 and -3
with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.
Then $\mathbf{A} = \mathbf{PDP}^{-1} = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ 2 & 1 \end{pmatrix}$
- (b) The eigenvalues are of opposite signs.
So the equilibrium point is a saddle point and is unstable.
- (c) The direction of the trajectories all start parallel to $\mathbf{v}_2$ towards the origin, and then move away from the origin parallel to $\mathbf{v}_1$.



3. (a) By substituting $y_1 = y$ and $y_2 = y'$, we have

$$\begin{cases} y_1' = & y_2 \\ y_2' = & -2y_1 - \frac{c}{4}y_2 \end{cases}$$

- (b) From the given ODE, we know that the mass of the ball $m = 4$, the spring constant $k = 8$, with damping constant c .
For under-damping, the condition is $c^2 < 4mk = 4 \times 4 \times 8 = 128$. So the possible values of c are $0 < c < \sqrt{128}$.

- (c) When $c = 8$, the underlying matrix of the system is $\begin{pmatrix} 0 & 1 \\ -2 & -2 \end{pmatrix}$.

In this case, the eigenvalues are the roots of $\lambda(\lambda + 2) + 2 = \lambda^2 + 2\lambda + 2$, which are given by the complex conjugates: $-1 \pm i$.

To find an eigenvector of $-1 + i$, we solve

$$\begin{pmatrix} -1+i & -1 \\ 2 & 1+i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

As usual, we just need to solve one of the two equations in this system, say $(-1 + i)x - y = 0$.

By letting $x = t$, we have $y = (-1 + i)t$. So by setting $t = 1$, we have the eigenvector $\begin{pmatrix} 1 \\ -1+i \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} i$.

So a fundamental set of solutions of the system is given by

$$\mathbf{x}_{Re} = e^{-t} \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos t - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin t \right)$$

$$\mathbf{x}_{Im} = e^{-t} \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} \sin t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos t \right)$$

The general solution of the system is:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 e^{-t} \begin{pmatrix} \cos t \\ -\cos t - \sin t \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \sin t \\ -\sin t + \cos t \end{pmatrix}$$

4. (a) By MATLAB, we see that the matrix $\mathbf{A}$ has a pair of complex eigenvalues $-2 + 4i, -2 - 4i$ and corresponding eigenvectors $\begin{pmatrix} 2 \\ -1+2i \end{pmatrix}, \begin{pmatrix} 2 \\ -1-2i \end{pmatrix}$.

So a fundamental set of real solutions:

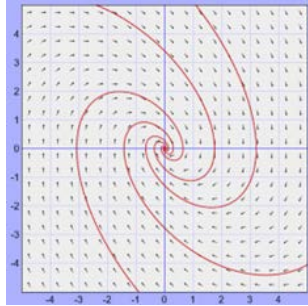
$$X_{Re} = e^{-2t} \left[\begin{pmatrix} 2 \\ -1 \end{pmatrix} \cos 4t - \begin{pmatrix} 0 \\ 2 \end{pmatrix} \sin 4t \right]$$

$$X_{Im} = e^{-2t} \left[\begin{pmatrix} 2 \\ -1 \end{pmatrix} \sin 4t + \begin{pmatrix} 0 \\ 2 \end{pmatrix} \cos 4t \right]$$

- (b) Since the eigenvalues are complex with negative real part, the equilibrium point is a spiral point and is asymptotically stable.

- (c) Test point: $\begin{pmatrix} 0 & 4 \\ -5 & -4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \end{pmatrix}$ (downward).

So the spiral is in clockwise direction spiralling towards the origin.



5. (a) ODE: $y'' + cy' + 4y = 0$ convert to
 SDE: $\begin{cases} y_1' = y_2 \\ y_2' = -4y_1 - cy_2 \end{cases}$ or $\mathbf{y}' = \begin{pmatrix} 0 & 1 \\ -4 & -c \end{pmatrix} \mathbf{y}$.
- (b) For degenerate node, $c^2 = 4mk = 4 \times 1 \times 4 \Rightarrow c = 4$.
- (c) By MATLAB, $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -4 & -4 \end{pmatrix}$ has a repeated eigenvalue -2 with only one linearly independent eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.
 We need to find a generalised eigenvector:
 $\left(\begin{array}{cc|c} 2 & 1 & 1 \\ -4 & -2 & -2 \end{array} \right) \Rightarrow x = s, y = 1 - 2s$.
 Set $s = 0$ we get a generalised eigenvector $\mathbf{u} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
 So the general solution of the SDE:
 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-2t} + c_2 \left[\begin{pmatrix} 1 \\ -2 \end{pmatrix} t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] e^{-2t}$ or
 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} t \\ -2t + 1 \end{pmatrix} e^{-2t}$.
- (d) IVP: $y_1(0) = 0.5, y_2(0) = 0.5$.
 $\begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
 Solve $c_1 = 0.5$ and $-2c_1 + c_2 = 0.5 \Rightarrow c_2 = 1.5$.
 So $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 0.5 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-2t} + 1.5 \begin{pmatrix} t \\ -2t + 1 \end{pmatrix} e^{-2t}$.
 Hence the solution of the ODE is given by:
 $y(t) = 0.5e^{-2t} + 1.5te^{-2t} = e^{-2t}(0.5 + 1.5t)$.
6. (a) (A) Saddle point; unstable (the direction field suggests all trajectories have the saddle shape with both tails far away from origin.)
 (B) Centre; neutrally stable (the direction field suggests all trajectories are concentric close loops around the origin.)
 (C) Spiral; unstable (the direction field suggests all trajectories are spiraling outwards away from the origin.)
 (D) (Star) node; asymptotically stable (the direction field suggests all trajectories are straight lines moving towards the origin.)

- (b) The underlying matrix of the SDE is $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ with eigenvalues $\lambda = \pm i$.

This suggests that the equilibrium point is a centre. So it corresponds to the direction field (B).

7. (a) To find a generalised eigenvector corresponding to $\mathbf{v}$, we solve

$$(\mathbf{A} + 2\mathbf{I})\mathbf{u} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

On solving, we get $x = \frac{1}{3}$ and $y = s$ where s is a parameter.

So we can take the generalised eigenvector to be $\mathbf{u} = \begin{pmatrix} \frac{1}{3} \\ s \end{pmatrix}$ for any s .

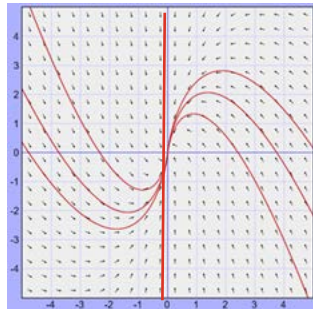
- (b) Let's take s in the generalised eigenvector be $s = 0$. Then the general solution of the SDE is given by:

$$\mathbf{y} = c_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} \frac{1}{3} \\ t \end{pmatrix} e^{-2t}.$$

- (c) The phase portrait of a degenerate node with axis along the eigenvector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is as shown below.

The direction of the trajectories can be determined by testing the point $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$\begin{pmatrix} -2 & 0 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \text{ which is a tangent pointing upward.}$$



- (d) Substituting the initial values, we get:

$$c_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

This gives $c_1 = 1$ and $c_2 = 3$.

So the particular solution for the IVP is given by:

$$\mathbf{y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-2t} + 3 \begin{pmatrix} \frac{1}{3} \\ t \end{pmatrix} e^{-2t} = \begin{pmatrix} 1 \\ 1 + 3t \end{pmatrix} e^{-2t}.$$

8. (I) Eigenvalues of opposite sign. This is a saddle point, and it corresponds to phase portrait B.
 (II) Eigenvalues distinct and both positive. This is an unstable node, and it corresponds to phase portrait A.
 (III) Eigenvalues complex without real parts. This is a center, and it corresponds to phase portrait F.

- (IV) There is a 0 eigenvalue. This is the special case, and it corresponds to phase portrait E.
- (V) Eigenvalue is repeated with only one linearly independent eigenvector. This is a degenerate node, and it corresponds to phase portrait C.
- (VI) Eigenvalues complex with real parts. This is a spiral point, and it corresponds to phase portrait D.